

Filipe Nunes Vicente

Research Scientist in Cell Biology and Imaging

Address: Felix-Wankel Strasse 19, 69126 Heidelberg, Germany

Email: filipenunesvicente1@gmail.com | **Contact:** +33695702853 | **ORCID:** 0000-0001-5107-0461

Linkedin: filipe-nunes-vicente | **GitHub:** fnunesv

SUMMARY

I am an interdisciplinary cell biologist and biophysicist with 10 years of experience in cellular and molecular biology, advanced fluorescence microscopy, and the development of quantitative assays. I am passionate about designing synthetic molecular tools to regulate cellular functions and developing high-throughput imaging methods.

EXPERIENCE

Postdoctoral researcher

01/2022-Present

Diz-Muñoz Lab, EMBL Heidelberg, Germany

- Developed three main GLP-compliant workflows for engineering multiclonal and single clonal cell lines:
 - 3T3 fibroblasts expressing inducible or constitutive custom-designed optogenetic actin crosslinkers coupled to signalling biosensors or tagged cytoskeletal proteins.
 - MEF fibroblasts with CRISPR knockouts for the membrane skeletal protein spectrin.
 - MDCK epithelial cells and W256 carcinosarcoma cells expressing inducible optogenetic actin crosslinkers.
- Designed six different optogenetic CRY2 or iLID-based actin crosslinkers with scalable molecular lengths, enabling nanometric control of actin spacing from 10 to 50nm.
- Performed molecular and phenotypic characterisation of engineered cell lines through PCR, Western Blot, flow cytometry, bulk RNA-seq, and automated live cell imaging.
- Established 3D cell culture and imaging of MDCK epithelial multicellular cysts up to 10 days of maturation.
- Designed a magnetic bead assay coupled to optogenetics to measure material properties of live cells with nanometric precision and time resolution of ~1s.
- Developed and implemented three high-throughput and machine learning workflows for microscopy:
 - Feedback Airyscan microscopy 2D and 3D timelapses coupled to single-cell optogenetic activation of up to 20 cells per sample.
 - Automated Nikon SoRA confocal spinning disk microscopy for 3D imaging of cytoskeletal proteins in hundreds of fixed cells per sample.
 - Four AI-driven image analysis pipelines to segment, quantify and classify microscopy data for more than 20 different morphodynamic parameters.
- Supervised two MSc students and collaborated with international teams of physicists, biologists, and chemists.
- Presented work in several interdisciplinary international conferences and partner labs.

Predoctoral researcher and scientist

09/2016-09/2021

Giannone Lab, IINS Bordeaux, France

- Developed two custom-microfabricated 3D-printed devices to combine for the first time single and cyclic cell stretching with super-resolution microscopy and single particle tracking.
- Implemented long-term sterile culture of primary hippocampal neurons and chicken spinal cord explants in custom-microfabricated stretching devices.
- Applied stretching devices to study and model biological processes in various cell types expressing fluorescent-tagged target proteins:
 - Cell-matrix adhesion and cytoskeletal organization in fibroblasts (MEF).
 - Membrane structures in endothelial cells (MLEC).
 - Axonal injury in primary neurons and chicken spinal cord explants.
- Optimized click-chemistry probes for super-resolution microscopy of dendritic spines in primary neurons.
- Published two first-author articles in high-impact-factor journals and collaborated with four other publications.
- Supervised one MSc and one PhD project on neuronal stretching with super-resolution microscopy.
- Collaborated with interdisciplinary research teams in physics, optics, and chemistry.
- Presented work in several interdisciplinary international conferences and partner labs.

GRANTS AND AWARDS

- Joachim Herz Add-On Fellowship for Interdisciplinary Life Sciences: 12500 € (01/2023-01/2026)
- EMBO Postdoctoral Fellowship: full postdoctoral salary coverage (01/2022-01/2024)
- French MCESR PhD grant: PhD salary coverage (09/2016-09/2019)

EDUCATION

University of Bordeaux, France **09/2016-11/2020**

PhD in Cell Biology and Neuroscience (Thesis supervisor: Dr. Grégory Giannone)

University of Coimbra (Portugal) /Neurocentre Magendie(France) **09/2014-07/2016**

MSc. in Cellular and Molecular Biology (MSc supervisors: Giovanni Marsicano and Román Serrat)

University of Coimbra (Portugal) **09/2011-07/2014**

B.Sc. in Biochemistry

SKILLS:

- **Cell and tissue culture:** adherent cell lines (fibroblasts, epithelial, endothelial); suspension cell lines (carcinosarcoma), primary hippocampal neurons; tissues (chicken spinal cord explants).
- **Cellular and molecular biology:** stable cell lines and single cell clones; transfection (electroporation, lentiviral infection, lipofection); flow cytometry; immunocytochemistry; CRISPR knockouts; plasmid design (Gibson Assembly, site-directed mutagenesis, PCR-based cloning); Western Blot.
- **Optical microscopy:** super-resolution widefield and confocal (PALM, STORM, DNA-PAINT, STED); single particle tracking; confocal spinning-disk and AiryScan; phase contrast; adaptive feedback microscopy.
- **Image analysis:** machine learning image analysis and segmentation (Fiji, CellPose, Napari, Ilastik); cell and single molecule tracking (Imaris, Metamorph, TrackMate); multidimensional image processing and visualization.
- **Material science and bioengineering:** Atomic force microscopy; magnetic-bead rheology; microfabrication; 3D-printing and milling; 3D cell culture and elastic substrates (collagen gels, PDMS assemblies); micropatterning.
- **Programming:** R (advanced); Python (intermediate); ImageJ macro language (proficient); Claude Code; Git.
- **Software:** image edition (Illustrator, Photoshop); computer-assisted design (FreeCAD, Solidworks), imaging acquisition software (NIS Elements, LAS X, ZEN, Metamorph); flow cytometry (FlowJo); ELN (eLabJournal).

SELECTED PUBLICATIONS

Nunes Vicente, F. *et al.* 2022. Molecular organization and mechanics of single vimentin filaments revealed by super-resolution imaging. *Science Advances* 8. <https://doi.org/10.1126/sciadv.abm2696>.

Nunes Vicente, F.* *et al.* 2020. Cell stretching is amplified by active actin remodelling to deform and recruit proteins in mechanosensitive structures. *Nature Cell Biology* 22. <https://doi.org/10.1038/s41556-020-0548-2>

PROFESSIONAL TRAINING

Lab Leadership for Scientists, EMBO, Heidelberg, Germany **11/2024**

Project Management for Scientists, EMBO (online) **10/2024**

EMBO Practical Course in Computational Optical Biology, IGC Lisbon **10/2022**

COMMUNITY INVOLVEMENT AND VOLUNTEERING

Translator, Scientific outreach, EMBL Heidelberg, Germany (04/2026) – High school resources on epigenetics

Practical Course Instructor, EMBL Heidelberg, Germany, (09/2024;09/2022) - EMBO Current Methods in Cell Biology (AFM, micropatterning)

Member, CoMaking Space, Heidelberg, Germany, 01/2022-Present – 3D-printing, woodworking

Member, Graines de Solidarité, Bordeaux, 09/2020-09/2021 - Non-profit organisation for food poverty

French tutor, Portuguese Consulate, Bordeaux, 09/2019 – 09/2021

Practical Course Instructor, University of Bordeaux, 09/2018- Cajal Neuroscience Summer School: super-resolution microscopy in neurons

Trainee Journalist, Rádio Universidade de Coimbra, 09/2014-09/2015 - Hosting live broadcasts and interviews, news redaction, creation of podcasts.

LANGUAGES

Portuguese (native); English (proficient); French (advanced); Spanish (advanced); German (elementary)